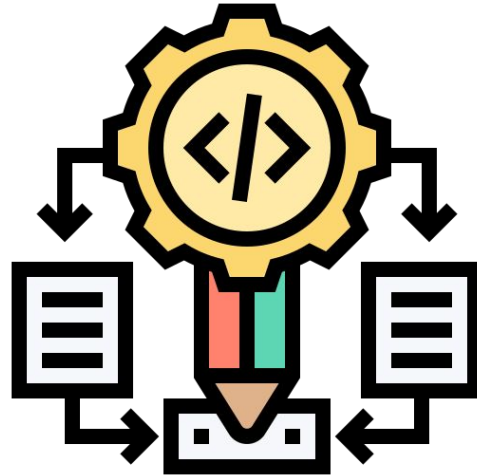


# Algorithm 1



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# Structs

بنیات

# Structures (Structs)

- A way to group several related variables into one place
- Each variable in the structure is known as a **member** of the structure
- Example

```
struct MyStructure {    // Structure declaration
    int myNum;           // Member (int variable)
    char myLetter;       // Member (char variable)
}; // End the structure with a semicolon
```

# Structures (Structs)

| Algorithm Syntax                                                                                                                                                 | C Syntax                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>Algorithm Example Begin   type Person = record     name : string     age : integer   end   var p : Person   p.name := 'Ibn Taymiya'   p.age := 30 End</pre> | <pre>#include &lt;stdio.h&gt;  struct Person {   char name[50];   int age; };  int main() {   struct Person p;   strcpy(p.name, "Ibn Taymiya");   p.age = 30;   return 0; }</pre> |

# Structures (Structs)

```
#include <stdio.h>
struct myStructure {
    int myNum;
    char myLetter;
};
void main() {
    struct myStructure s1;
    // Assign values to members of s1
    s1.myNum = 13;
    s1.myLetter = 'B';

    // Print values
    printf("My number: %d\n", s1.myNum);
    printf("My letter: %c\n", s1.myLetter);
}
```



```
My number: 13
My letter: B
```

# Structures (Structs)

```
void main() {  
    // Create an array of structures with 3 elements  
    struct myStructure arr[3];  
    // Initialize each element of the array  
    arr[0].myNum = 100;  
    arr[0].myLetter = 'A';  
    arr[1].myNum = 200;  
    arr[1].myLetter = 'B';  
    arr[2].myNum = 300;  
    arr[2].myLetter = 'C';  
    // Display the values stored in the array  
    for (int i = 0; i < 3; i++) {  
        printf("%d,%c\n", arr[i].myNum, arr[i].myLetter);  
    }  
}
```



```
100,A  
200,B  
300,C
```